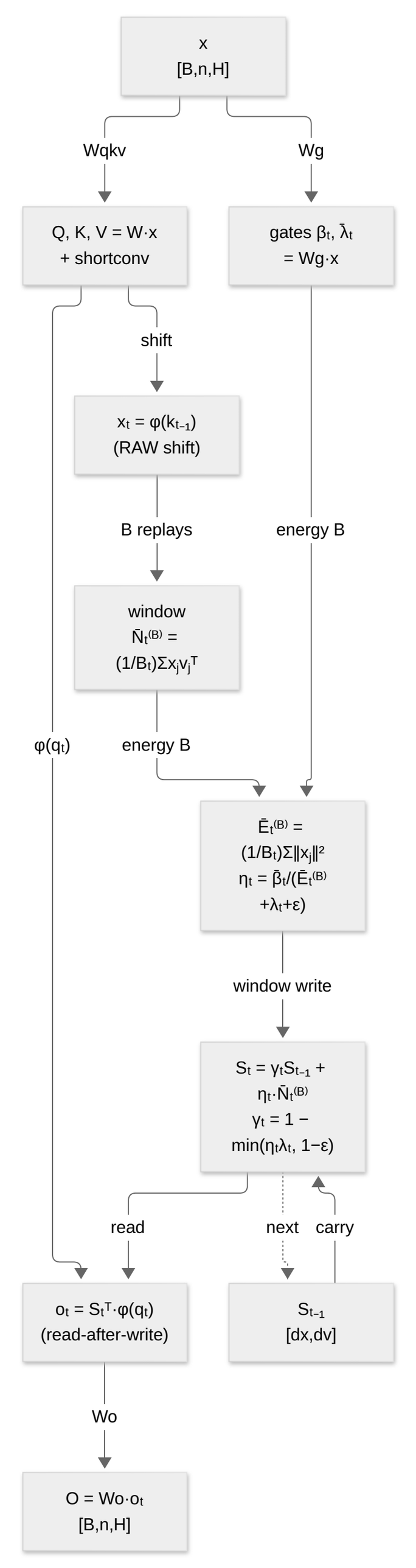

x
[B,n,H]
Wqkv
Wg
Q, K, V = W·x
+ shortconv
gates β_t, $\bar{\lambda}_t$
= Wg·x
shift
$x_t = \varphi(k_{t-1})$
(RAW shift)
B replays
energy B
window
$\bar{N}_t^{(B)} =$
$(1/B_t)\Sigma x_j v_j^T$
$\varphi(q_t)$
energy B
$\bar{E}_t^{(B)} =$
$(1/B_t)\Sigma\|x_j\|^2$
$\eta_t = \bar{\beta}_t/(\bar{E}_t^{(B)}$
$+\lambda_t+\varepsilon)$
window write
$S_t = \gamma_t S_{t-1} +$
$\eta_t \cdot \bar{N}_t^{(B)}$
$\gamma_t = 1 -$
$\min(\eta_t\lambda_t, 1-\varepsilon)$
read
next
carry
$o_t = S_t^T \cdot \varphi(q_t)$
(read-after-write)
S_{t-1}
[dx,dv]
Wo
O = Wo·o_t
[B,n,H]